

Project Sunfish revival

<https://bluerobotics.com/learn/bluerov2-assembly/#introduction>

5-28-26

- Found laptop passwords
- Took apart electronics inside enclosure and sorted components into different piles
- Cleaned inside the battery enclosure and re-greased o-rings in all acrylic tubes.
- Tore down ROV2 to the “factory” setup
- Identified components added from OG project
 - GPS sensor in main electronics tube
 - Buck Converter attached to main battery to get power and main board housing inside main tube
 - Vernier optical dissolved oxygen probe with *complete* setup
- Added missing structural bolts to the frame and tube clamps themselves

5-29-26

- Start to put the main board back together
- Removed GPS/compass from pixhawk

What to do

Use the **ArduSub/BlueROV2 motor test** and verify one motor at a time:

1. Remove propellers for safety.
2. Open **BlueOS / Cockpit** or **QGroundControl** motor test.
3. Test **Motor 1**.
4. See which physical thruster moves.
5. If the wrong one moves, move that ESC signal plug to the correct **MAIN OUT** port.
6. Repeat for every thruster.

Simple rule

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MAIN OUT 1 → Motor/Thruster 1
MAIN OUT 2 → Motor/Thruster 2
MAIN OUT 3 → Motor/Thruster 3
...
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- For your BlueROV2 don't just plug them in randomly. Match the ESC signal leads to the **motor numbering**
- Need to do motor test next week
- Replaced water sensors, The “sponge” in the enclosure was expanded meaning there was moisture collected, either from long term storage and the sponge collecting moisture from the air, or actual water introduced to the system. I am more confident that it was moisture pulled from the air due to the storage factors.
- Replaced communication wire on the moisture sensor board as a precaution.
- The board is probably 75% completed!

6-1-26

- Connected the rest of the board and put together the enclosure

6-2-26

- Re-wired the inputs for the Pixhawk
- Added the remaining 2 ESC units and wired them to the 2 thrusters left
- Ensured all connections were tight, re-soldered and shortened some cables.
- Performed vacuum test number 1 : Passed!
- Performed vacuum test number 2 : Passed!
- Removed battery from chassis and placed on charger
- Found issue with tether connection, doesn't connect to itself so we don't know how we can control it.

6-3-26

- Took apart enclosure and added ethernet cable connections, and connection from pi to pixhawk
- Performed vacuum test number 3 : Passed
- Pulled apart board and drew board diagram
- Identified potential issue with tether connection

6-4-26

- Found tether issue
- Made tether pin layout/diagram with connections
- Put board back together and performed vacuum test 4 : Passed
- Connected depth sensor
- Performed vacuum test 5 : Passed
- Set up ground control on Ethan's laptop
- Went through sensor setup
- ROV2 is ready to test in water, test date is thursday june 11th at 6:00am

Relevant documentation

https://bluerobotics.com/wp-content/uploads/2019/02/BlueROV2_heavy.pdf Blue rov2 connections

<https://www.ardusub.com/introduction/hardware-options/connection-diagrams.html> Connection diagrams

<https://bluerobotics.com/learn/bluerov2-software-setup/#installing-qgroundcontrol> ground control setup

<https://www.ardusub.com/introduction/hardware-options/required-hardware/tether.html> hardware setup with tether

<https://bluerobotics.com/learn/wl-connector-standard/#tether> standard connection

6-5-26

- Logo design

6-8-26

- Performed vacuum test number 6 : Passed
- Plugged vent holes
- Researched plug for loose end connector, best option might be to 3d print one and throw an o-ring in
- Pretty sure I identified the random cable. I believe it is the power/data cable for the doppler velocity log.
- I identified that the manufacturer did not intend the drone to be operated without the tether attached.
- SEACON Underwater Wet-Mate Connectors is the cable used
- The vga cable is used for the doppler sensor, can be removed
- I believe it would be a good idea to add the ping sonar that we currently own, it would add a "failsafe" if the camera feed from the drone fails so we wouldn't crash into anything

6-9-26

- Reviewed all of the flight logs that Amanda completed to determine how and when the camera failed
- Re-wired the camera, fixing the problem unless the camera is fried due to the failed ground wire from the previous use.
- Attempted to unscrew the back caps of the enclosure to add the sonar sensor
- Safely and securely wired sonar sensor into the BLUART converter that allows it to be converted to usb, and then plug into the pi
- Removed the entire serial cable to usb as well as the "extra" tether that is used to communicate to the doppler sensor.
- Re-cracked open the enclosure to adjust orientation so the pixhawk would be facing starboard, therefore the gyroscope would be accurate
- Removed buoyancy foam to remove the final add ons from the previous project
- Re-attached the skid plate for safe sonar mounting, there were no available mounting slots for the sonar that would be safe, so the skid plate is now on
- Zip-tied the cords out of the way for a more clean/unimpeded thruster ability.

6-10-26

- Plugged in Sunfish to battery and connected to QGC to test camera and see if sonar was working.
- Camera does not seem to be working still.

6-11-26

- Prepped for first dive, made sure calibration was good, and all sensors were plugged in
- Checked to make sure all thrusters were plugged in
- Booted up Qground control and tested xbox controller.
- Made a few tweaks with the sunfish and all of the wires and thrusters.

6-12-26

- First dive!!!
- Thruster number 2 does not turn when in water
- There is a grinding noise when in water when turning it to the right, possibly the thruster itself, maybe another thruster grinding on something while attempting to turn
- Sub is a little too heavy, have to remove some of the ballast weights
- There is a small leak in the main enclosure of the sub, after being submerged at different depths for a consistent 10-12 minutes, there was about 5 tablespoons of water in the enclosure, nothing touched the electronics, only the wires, and everything from the outward view is still operational.
- I'm pretty sure the leak was due to a small gap between the cap and the enclosure as well as not lubricating the o-rings with silicon before the dive.
- Need to get more familiar with the controls as well as possibly investing in a wireless xbox controller, the wired one made it decently hard to navigate to a good vantage point while connected to the laptop
- Need to get a better look at the screen while diving to see what controls i can see and sensors are working, i know on the blue robotics page, there is a new operating system where you can customize your home screen and put different readouts/panels in the main view, worth looking into
- Need to research new thrusters, a new thruster from blue robotics with an extra fin is \$240, there might be some in longterm storage that we can find.

Controller Functions

The default button layout is shown below:



- Above is the controller mapping
- I also don't know if the camera tilt works, I am pretty sure that i can test it outside of water though, The only part that we need might be the actual camera from blue robotics, which is \$134.40 without shipping domestic shipping is \$6.50, 2 day shipping is \$35.90

6-16-26

- Dive date!!!
- Sunfish did not pass the leak test, there was too severe of a leak to safely perform the test today. Karson and I replaced the tubes for the vacuum pump as they were leaking, as well as took off the enclosure lids on either side, both the domed one and the corded one. And inspected for any hairs or debris that caused the improper test. Could not find anything. I suspect that the leak is coming from some sort of crack in the acrylic tubing, there are many scratches on the tubing itself and that could cause a leak. There was also a small leak that looked like it was allowing water in the underside of the sub, in the same position as the last leak, so I am confused as to why this is still happening. When I return on Monday the 22nd of June, I will investigate further and try to find the issue.

6-22-26

- Took off the 12 inch acrylic tube that I thought had the leak and replaced it with the 18 inch acrylic tube. For the 18" tube I made a board mount like the one attached to the camera to make it more stable for the actual board itself
- Vacuum tested the 18" tube and the pump was saying there was another leak. It wasn't constant but it was also hard to get up to pressure, making me think that it is the actual vacuum pump that is the issue. There are some other vacuum pumps on amazon that are available, there is the style of the one we currently have for around 15-20 and the more robust one for 57 without any tubes i think. Will investigate more tomorrow

6-23-26

- Tested vacuum for 18" tube and still failed, next step is to replace battery housing
- 3D printed and modded different camera mounting systems, as well as created internal skeleton to help with the stability of the camera in general

6-24-26

- Performed testing on why the camera was not working, turns out its the actual camera itself, not the board. Need to look into buying a new tested camera
- Hooked up QGround Control to Sunfish and tried to map out what channel the lights are on and decided that the lights are indeed not working, either from frankenstaining the wires or bad wiring in general. Need to look into further
- Replaced battery housing enclosure
- Performed vacuum tests to ensure vacuum

6-25-26

- Pulled cap plugs out and re-greased and replaced some of the o rings on certain plugs.
- Opened the enclosure and un-seated wires, re-routed most of the wires to make it a cleaner fit inside the enclosure, no more rubbing wires, plus that allowed me to ensure things were connected properly as well as double check the thruster inputs
- Opened most cap o-rings and replaced grease in them, some used the bad type of grease, cleaned and replaced all

6-26-26

- Put enclosure back together with everything wired together, this both fixes the orientation of the pixhawk for the gps issue that we were having, as well as it allows me to ensure that the seals on the inside are tight
- Tested each individual tube with the vacuum pump, turns out i think that it is the connection between the pump itself, we found an stl file that we are printing currently to test the vacuum out on.
- Will try and update on monday

6-29-26

- Attempted to find and fix pressure leak, unsuccessful, will try again tomorrow

7-1-26

- Re sealed enclosure and connected all of the bulkhead connectors
- Small leak is still there, but definitely slower, I don't really know what it could be outside of that besides some connection issue, but i will figure that out on monday

7-9-26

- Put all dome mounted plugs back in with re-greased O-rings and torqued down the cap screws with an allen wrench, performed the first pressure test after re-assembly : PASSED!

7-10-26

- Took time lapse video of putting sunfish back together for social media, re-wired everything and seated everything back together. Performed Pressure test #2 : PASSED!

7-13-26

- Preparation for dive #2
- Dive #2 was a success!
- Thruster number one is still non operational, and therefore causes the sunfish to descend into water unevenly
- Thruster 2 looks like it is always on

7-20-26

- Work on developing the automatic water collection system
 - Idea: Have a 3D printed honeycomb cage around a scientific ziploc bag. The bag needs to have a relatively wide opening, about the width of a quarter. This is attached to the ring of the cage. Attached to the cage, there is a flip top lid with a servo motor attached on top, or on the side. There is a spring holding the lid closed with constant pressure. The servo motor will then open the flip lid and let water in, when it is full, after a certain amount of time, the motor will release and the spring will cause the lid to snap shut, locking the water in the bag. These need to be all individually working, and there needs to be 3 different mounted

collection systems. The bag will also need to be easily removable and replaceable without letting any contaminants come in contact with the sample

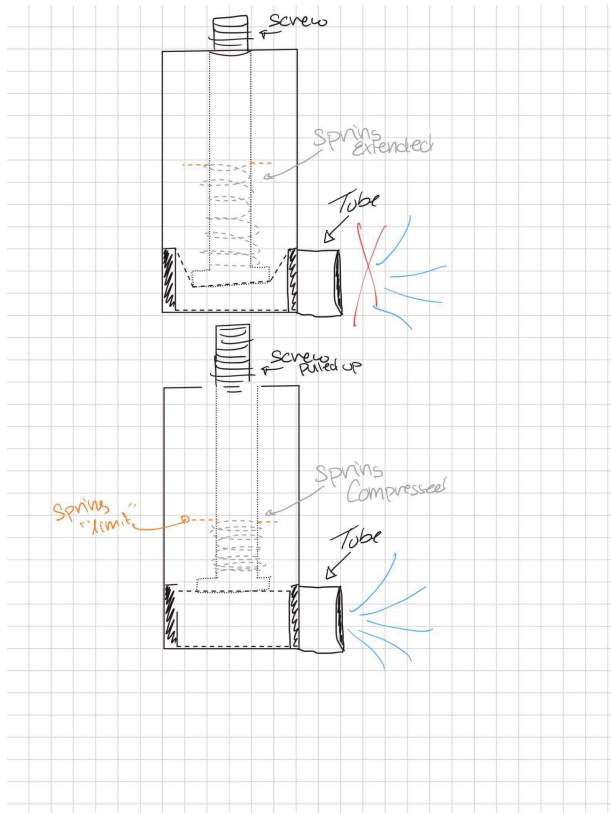
- 3D printed collection housing cage, and developed one idea for the lid
- Coded three servos to operate off one pi

7-21-26

- Modeled and Printed another design for the lid which allows a catheter type silicone tube to go through it and then be clamped off
- Coded a servo to clamp the tube, release the clamp to have the bag fill with water, and then clamp it again. The problem with this is if there is any movement in the servo itself, the positioning will be off and therefore not create a closure in the tube.
- Marz thought getting a different, more powerful servo would be a good idea, She is removing the one from the Boeing robot downstairs.

7-24-26

- With the help of Marz, figures out how to wire the dynamixels correctly and then also code them to starting position, 90 degrees closed, and then back to starting position. The positioning of the motors itself is repeatable and able to be produced more than once.
- Polished ideas for collection system, we will end up using a rubber duck bill valve to keep the water out, then the servo motor will apply pressure against it and then open it, when collection is complete, the servo will then remove pressure and seal the collection bag.



7-28-26

- Developed 2 prototypes of the sampling device, also developed and implemented a ball check valve to slip inside the collection tube, this will allow the bag to not be contaminated by any other samples.
- Started writing Github repository for the project.